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Introduction lecture

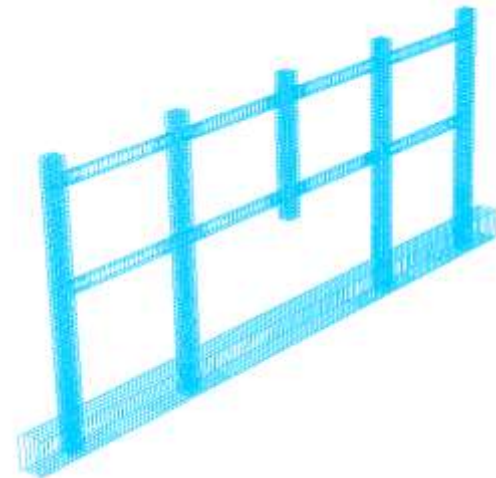
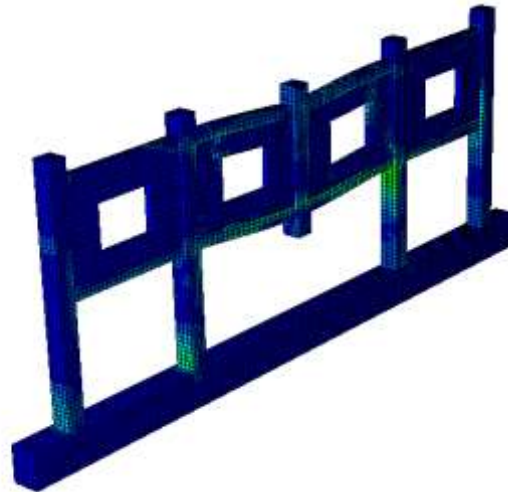
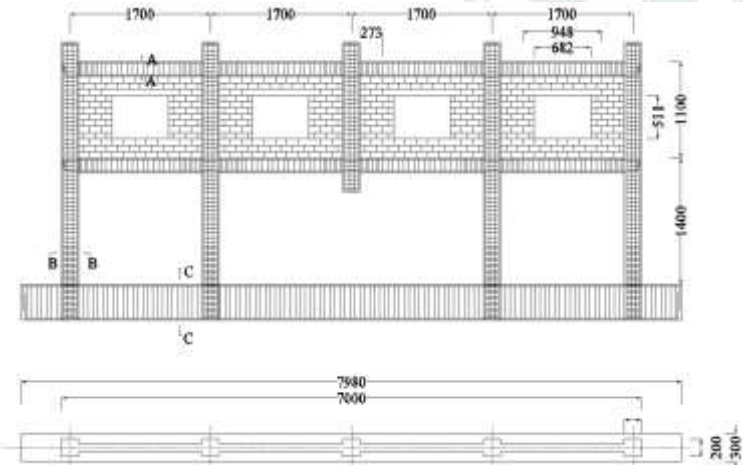
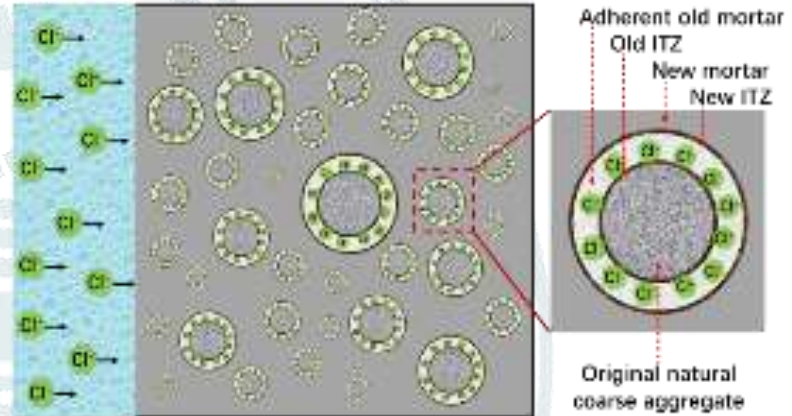
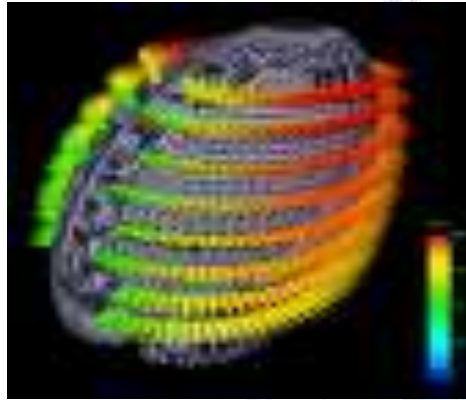
Prof. Jian Yang (杨健)
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What is FEM



FEM, also called as finite element analysis (FEA), is a method for numerical solution of field problems.





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1896

1920

1987

2006



落重冲击试验

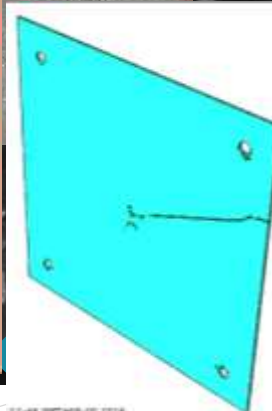
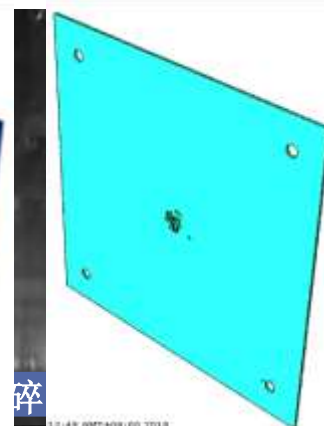
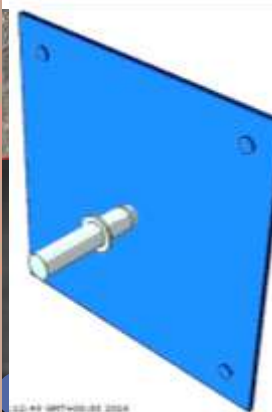
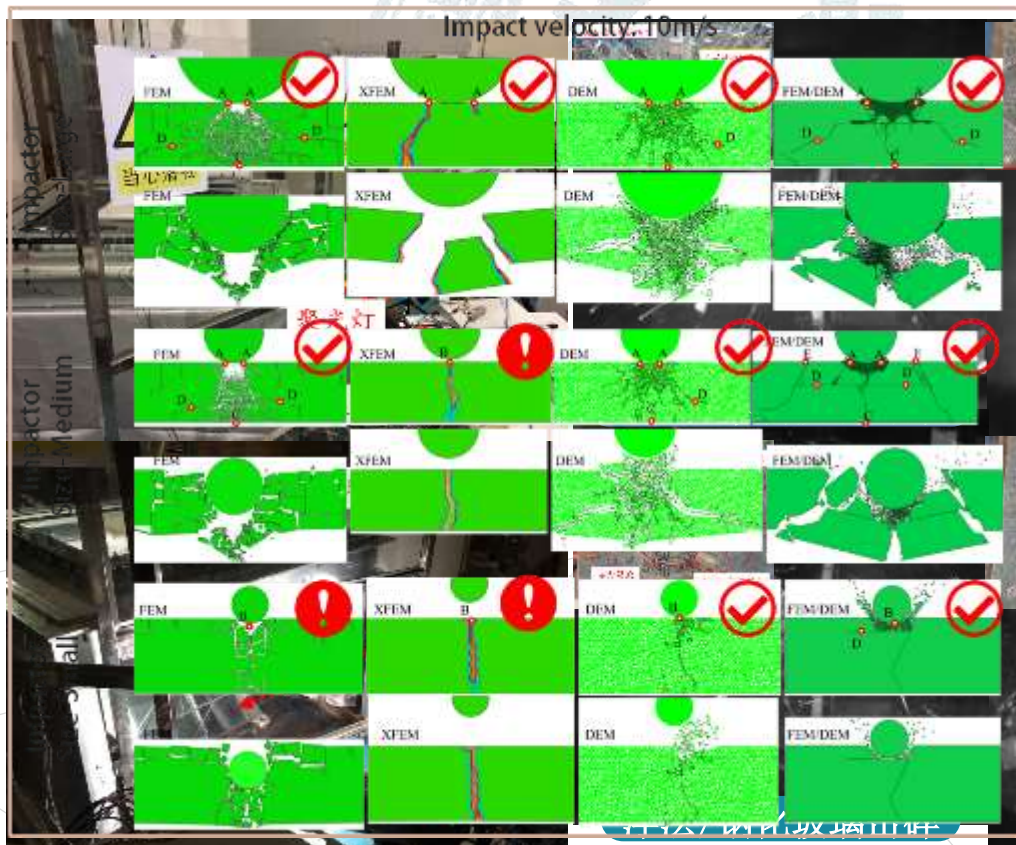
新型数值方法

FEM

XFEM

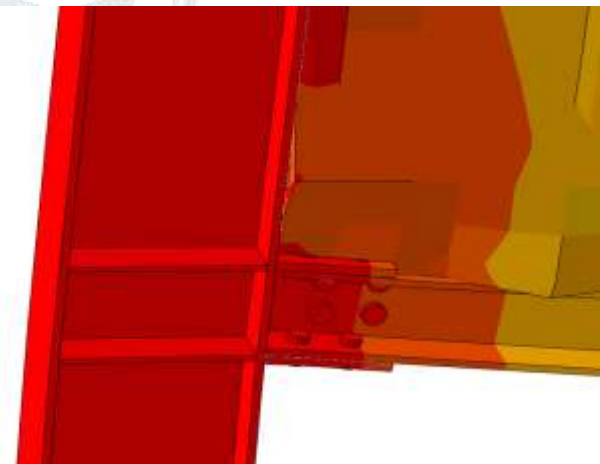
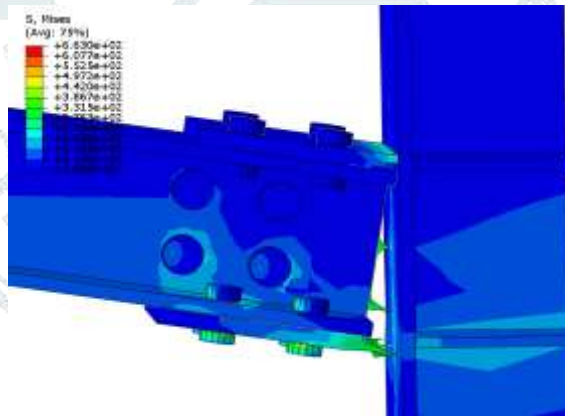
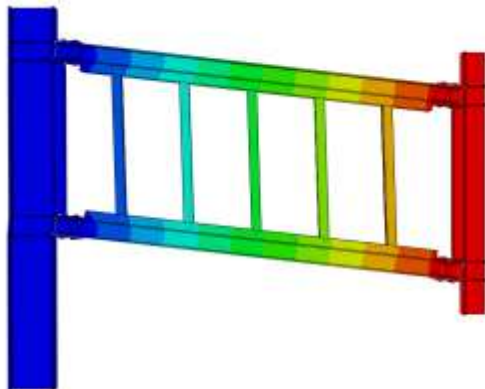
DEM

FEMDEM



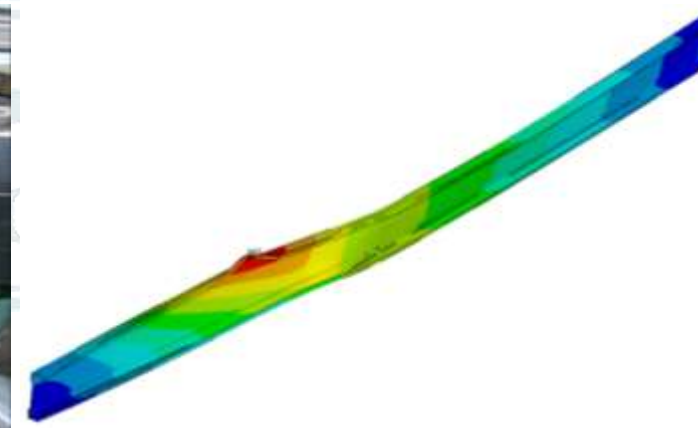
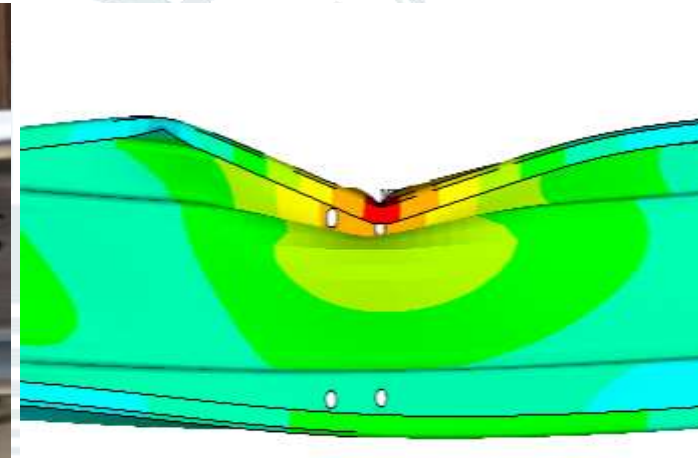
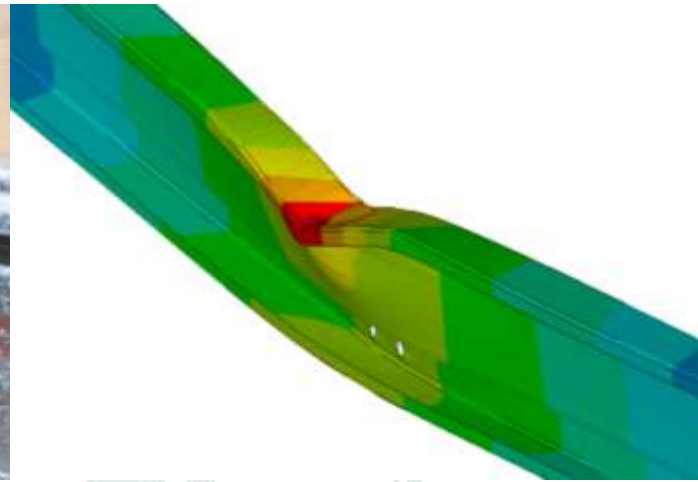
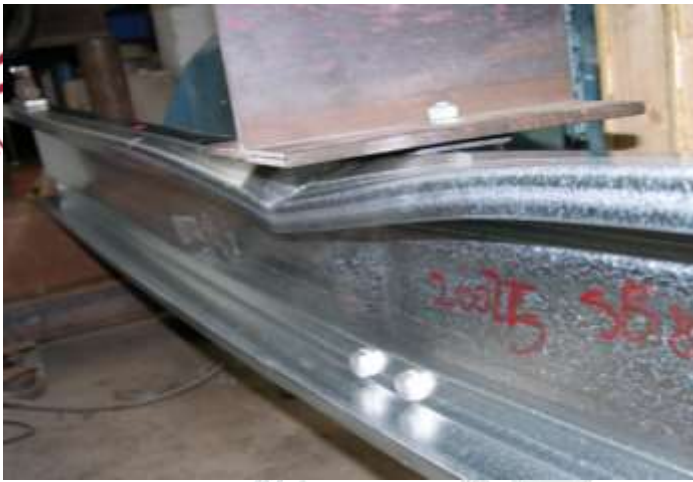


FE modelling





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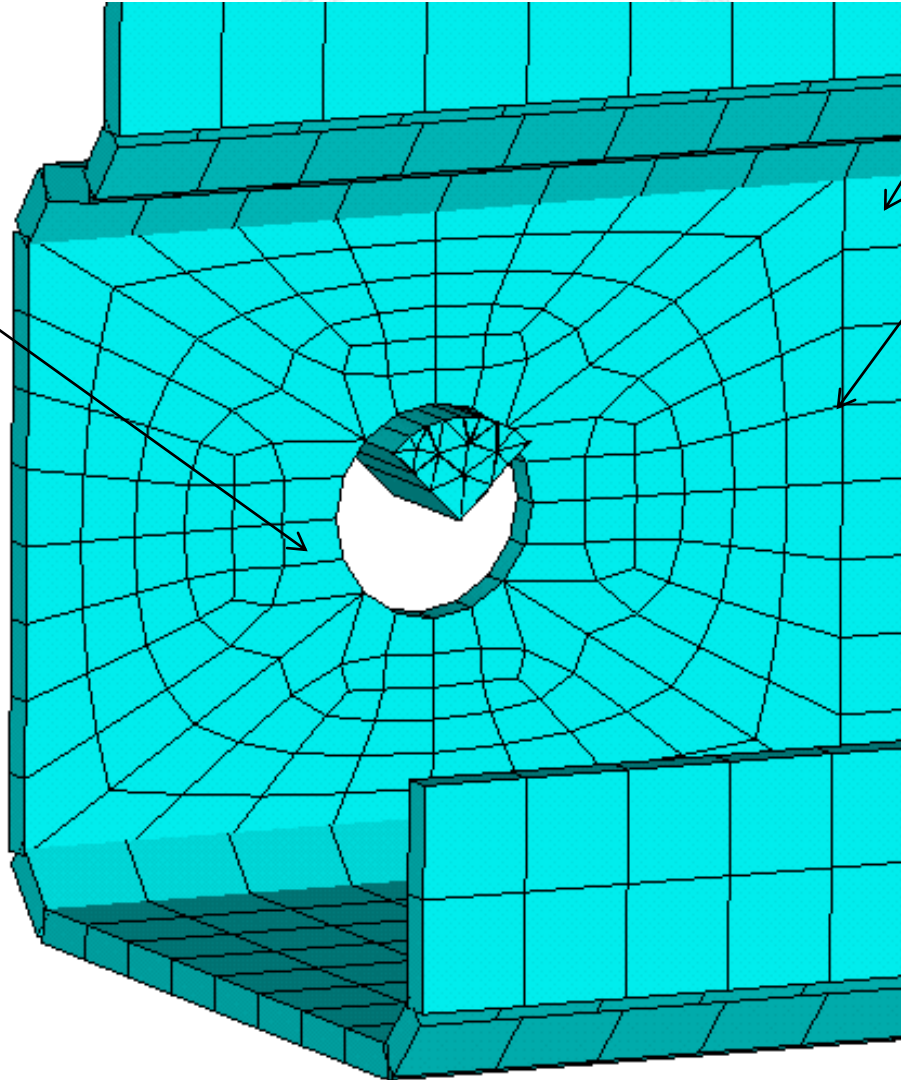
Discretisation



Element

Node

Mesh





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Alternative numerical methods

- Other numerical solution methods:
 - Finite differences
 - » Approximates the derivatives in the differential equation using difference equations.
 - » Useful for solving heat transfer and fluid mechanics problems.
 - » Works well for two-dimensional regions with boundaries parallel to the coordinate axes.
 - » Cumbersome when regions have curved boundaries.
 - Weighted residual methods (not confined to a small subdomain):
 - » Collocation
 - » Subdomain
 - » Least squares*
 - » Galerkin's method*
 - Variational Methods* (not confined to a small subdomain)



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Advantages of FEA

- ⊙ Versatile
- ⊙ No geometric restriction
- ⊙ Boundary conditions and loading are not restricted
- ⊙ Material properties are not restricted
- ⊙ Components that have different behaviours
- ⊙ An FE structures can closely resembles the actual body
- ⊙ The approximation is easily improved by grading the mesh or increasing the order of elements
- ⊙ Can deal with multi-physics problems.



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Terminologies

- Interpolation
- Elements
- Nodes
- D.O.F
- ○ ○ ○



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History of FEA

- 1943 – Courant (Variational method)
- 1956 (1954) – Turner, Clough, Martin and Topp (Stiffness)
- 1955 – Argyris
- Clough, Argyris, Zienkiewicz the three founders of FEA; K. Feng made great contributions but was less known.
- 1960– Clough (proposed Finite Element, plane problems)
- 1970s – Applications on mainframe computers
- 1980s – Microcomputers, pre- and post-processors
- 1990s – Analysis of large structural systems



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Available commercial FEM software packages

- ANSYS
- SDRC/I-DEAS (complete CAD/CAM/CAE package)
- Nastran
- ABAQUS
- COSMOS
- ALGOR
- PATRAN
- HyperMesh
- LS-Dyna
- Comsol



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Solving a problem by FEA

- Classification of problems
- Model idealization
- Preliminary analysis
- Finite element analysis
 - Preprocessing
 - Numerical analysis
 - Postprocessing
- Check and interpret results
- Reiteration

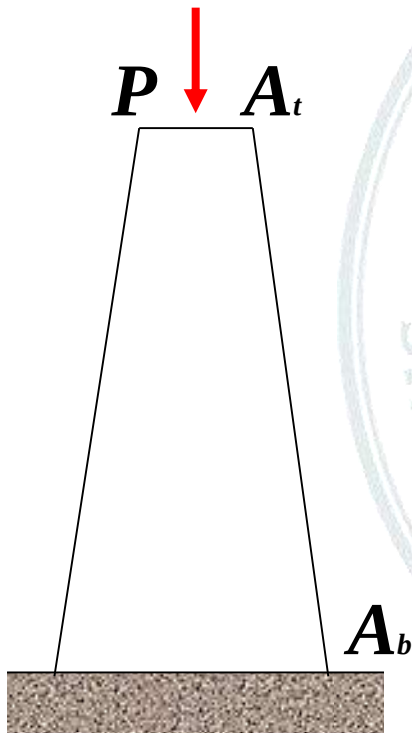


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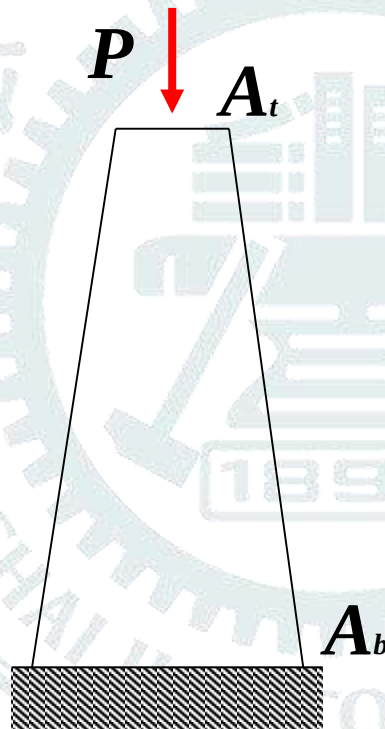


Structure



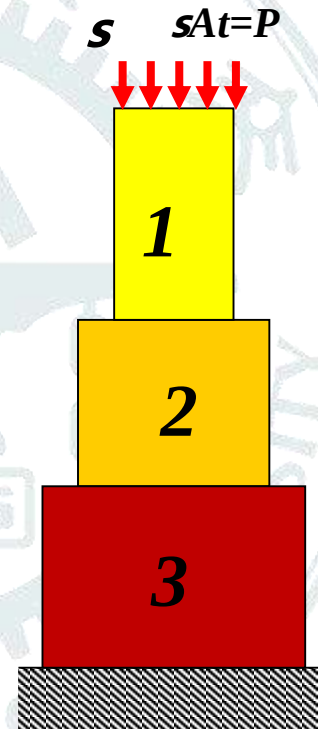
Ground

Model

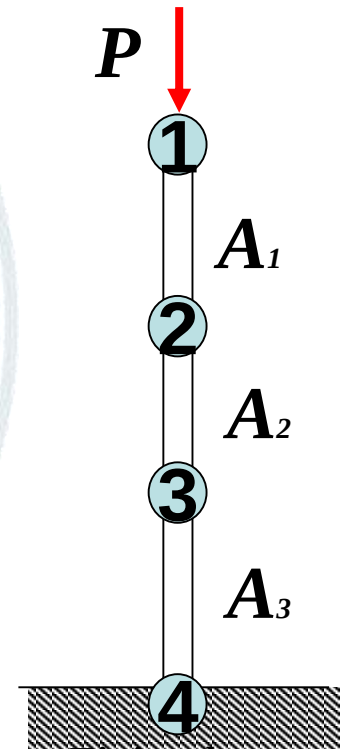


Rigid
support

Discretized models



Physical
representation



Finite element
representation



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Types of finite elements

- 1-D (line) element
- 2-D (plane) element
- 3-D (solid) element



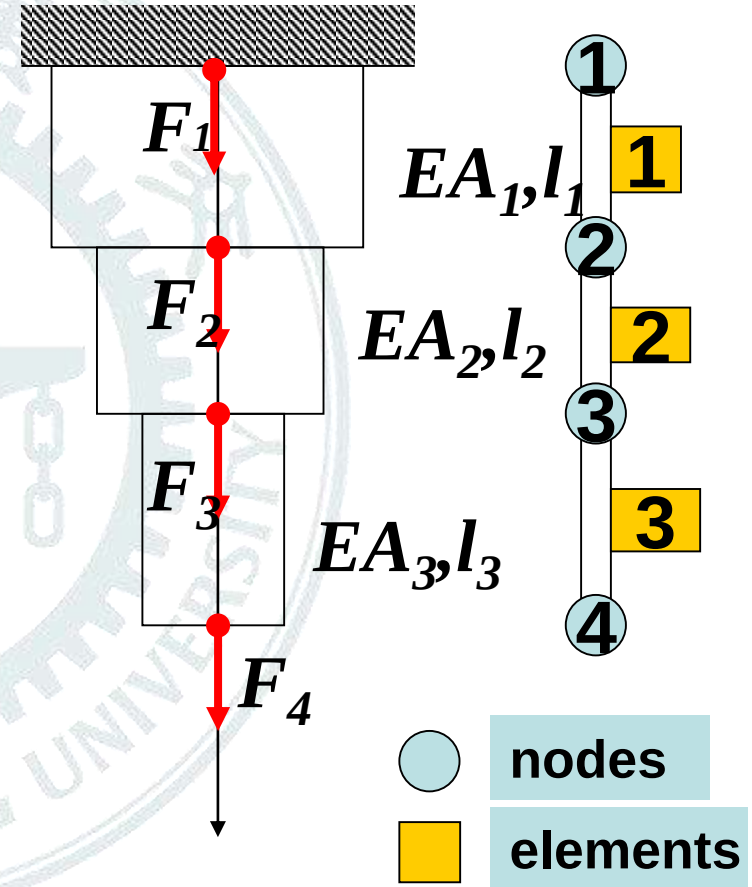
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How does the FEA work

Consider the bar of different cross-sections subjected to three point loads as shown in the figure. Determine the displacements at the loading points.





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Element stiffness matrices

Equilibrium equation
for element 1

$$\begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix} = \begin{Bmatrix} P_{11} \\ P_{12} \end{Bmatrix}$$

$$k_i = \frac{EA_i}{l_i}$$

Equilibrium equation
for element 2

$$\begin{bmatrix} k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{Bmatrix} u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} P_{21} \\ P_{22} \end{Bmatrix}$$

Equilibrium equation
for element 3

$$\begin{bmatrix} k_3 & -k_3 \\ -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} P_{31} \\ P_{32} \end{Bmatrix}$$

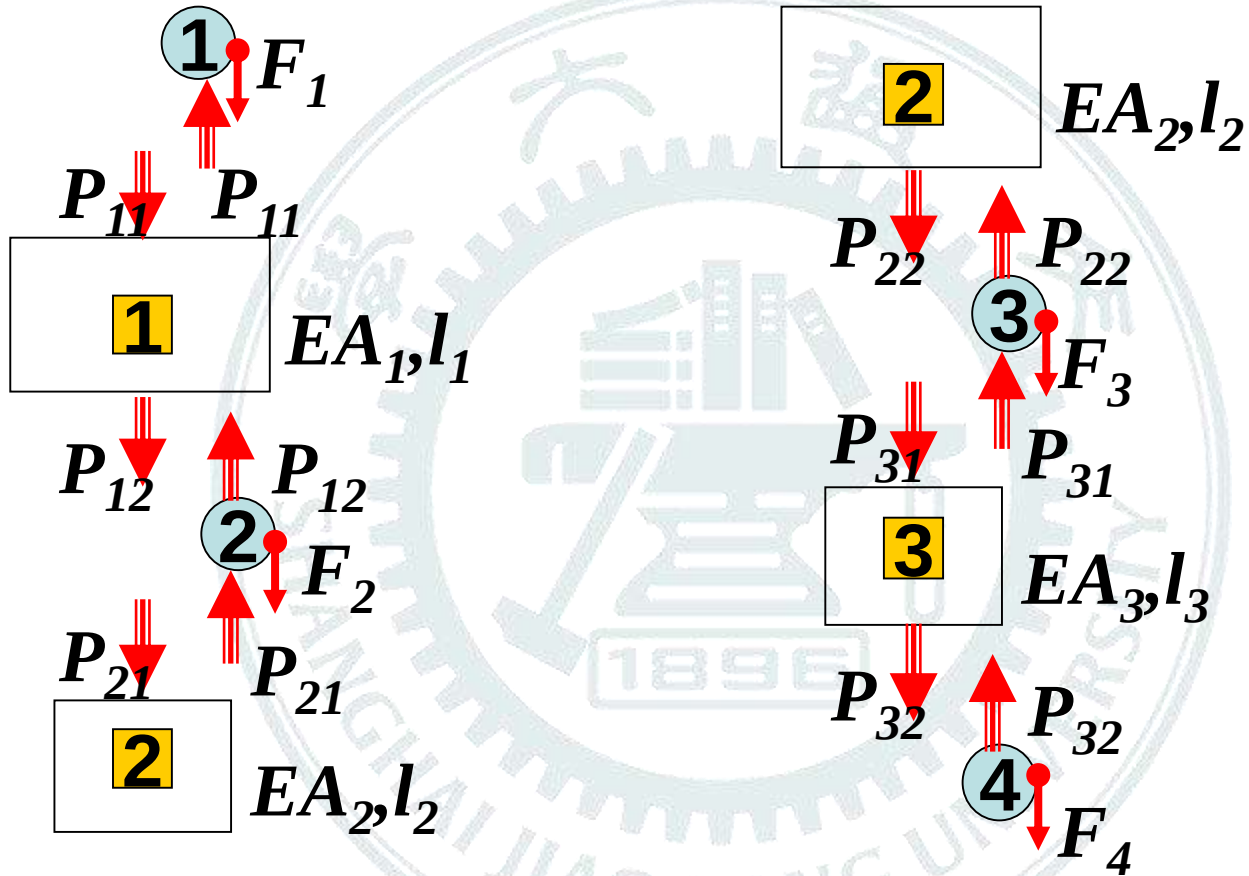


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Equilibrium of nodes



$$\begin{aligned} P_{11} &= F_1 \\ P_{12} + P_{21} &= F_2 \end{aligned}$$

$$\begin{aligned} P_{22} + P_{31} &= F_3 \\ P_{32} &= F_4 \end{aligned}$$



Assembly of stiffness matrices

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} P_{11} \\ P_{12} \\ 0 \\ 0 \end{Bmatrix} \quad \text{For element 1}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ P_{21} \\ P_{22} \\ 0 \end{Bmatrix} \quad \text{For element 2}$$

Element equilibrium equations in the global coordinate system



Assembly of stiffness matrices

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ P_{31} \\ P_{32} \end{Bmatrix} \quad \text{For element 3}$$

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} P_{11} \\ P_{12} + P_{21} \\ P_{22} + P_{31} \\ P_{32} \end{Bmatrix} = \begin{Bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$



Applying boundary conditions

$$u_1 = 0, \quad F_1 = R$$

Stiffness matrix

Global equilibrium equations

$$\begin{bmatrix} k_1 & -k_1 & 0 & 0 \\ -k_1 & k_1 + k_2 & -k_2 & 0 \\ 0 & -k_2 & k_2 + k_3 & -k_3 \\ 0 & 0 & -k_3 & k_3 \end{bmatrix} \begin{Bmatrix} 0 \\ u_2 \\ u_3 \\ u_4 \end{Bmatrix} = \begin{Bmatrix} R \\ F_2 \\ F_3 \\ F_4 \end{Bmatrix}$$

Solve for nodal displacements from the global equilibrium equations
Solve for reaction forces from the global equilibrium equations
Calculate the strain for each element from obtained displacements
Calculate the stress for each element from obtained displacements



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Properties of stiffness matrices

- ⦿ Nonnegative K_{ii}
- ⦿ Symmetry
- ⦿ Sparsity
- ⦿ Singularity (no support or mechanism)



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Checking results

- Deformed shape of the structure
- Balance of the external forces
- Order of magnitudes of results

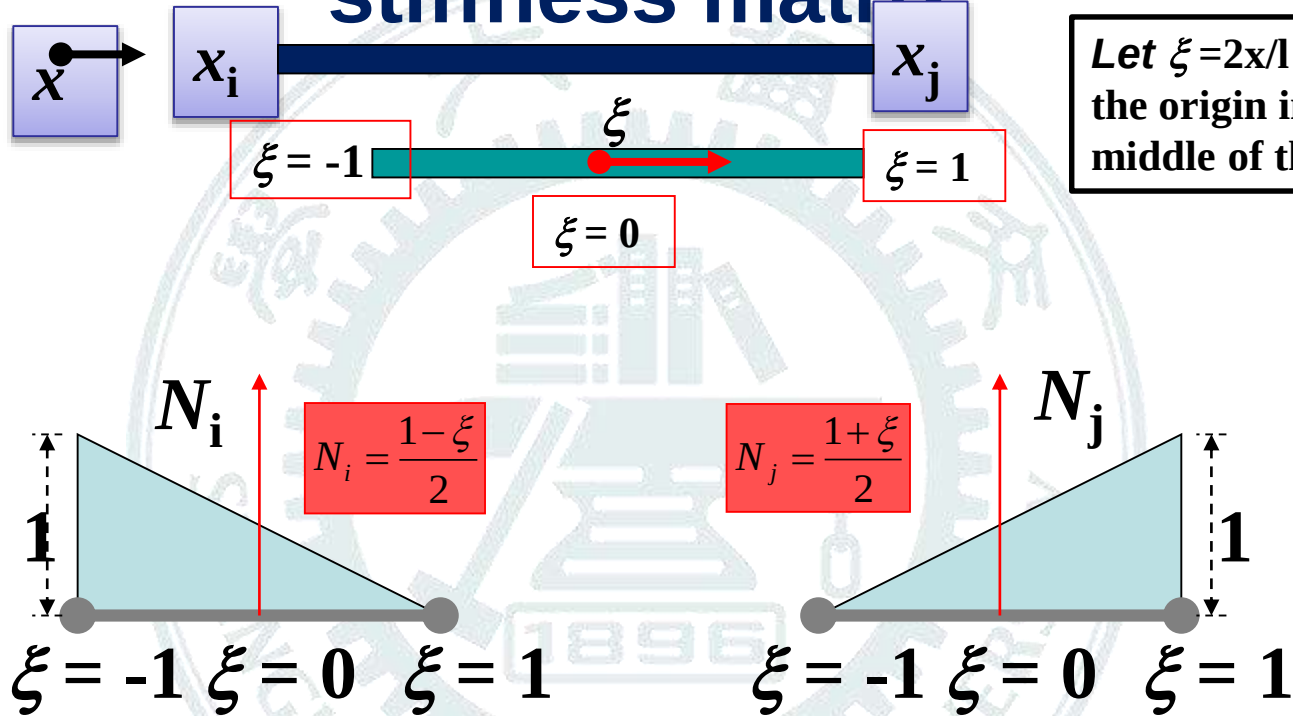


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A generalised approach to formulate stiffness matrix



$$x = N_i(\xi)x_i + N_j(\xi)x_j$$



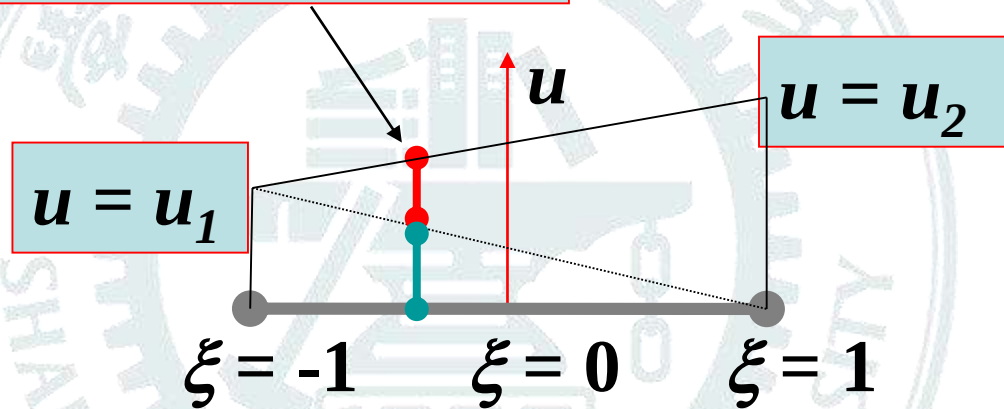
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Element displacement field

$$u(x) = N_i(\xi)u_1 + N_j(\xi)u_2$$



$$u = \begin{bmatrix} N_i & N_j \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$



Element strain field

$$\varepsilon(x) = \frac{du}{dx} = \begin{bmatrix} \frac{dN_i}{dx} & \frac{dN_j}{dx} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$\frac{dx}{d\xi} = \frac{dN_i}{d\xi} x_i + \frac{dN_j}{d\xi} x_j = -\frac{x_i}{2} + \frac{x_j}{2} = \frac{x_j - x_i}{2} = \frac{l}{2}$$

$$\frac{dN_i}{dx} = \frac{dN_i}{d\xi} \frac{d\xi}{dx} = \left(-\frac{1}{2}\right) \left(\frac{2}{l}\right)$$

$$\frac{dN_j}{dx} = \frac{dN_j}{d\xi} \frac{d\xi}{dx} = \left(\frac{1}{2}\right) \left(\frac{2}{l}\right)$$

$$\varepsilon = \begin{bmatrix} -\frac{1}{l} & \frac{1}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$



Element stress field

$$\sigma(x) = E\varepsilon(x) = E \begin{bmatrix} \frac{dN_i}{dx} & \frac{dN_j}{dx} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$\sigma = \begin{bmatrix} -\frac{E}{l} & \frac{E}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

Note: element strain and element stress are constant



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Element internal force

$$N(x) = \sigma(x)A = EA \begin{bmatrix} \frac{dN_i}{dx} & \frac{dN_j}{dx} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$N = \begin{bmatrix} -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$



Element strain energy

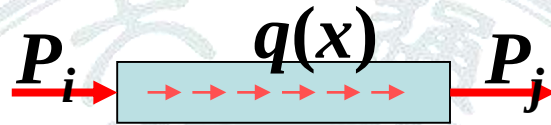
$$U = \frac{1}{2} \int_0^V \sigma \varepsilon dV = \frac{Al}{2} \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{bmatrix} -\frac{E}{l} \\ \frac{E}{l} \end{bmatrix} \begin{bmatrix} -\frac{1}{l} & \frac{1}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$
$$= \frac{1}{2} \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix}$$

$$[K] = \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix}$$

Element stiffness matrix



Work in the element



$$\begin{aligned} W &= -\int_l q u dx - (P_i u_i + P_j u_j) \\ &= -q \int_l \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} N_i \\ N_j \end{Bmatrix} dx - (P_i u_i + P_j u_j) \\ &= -\begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} \frac{ql}{2} \\ \frac{ql}{2} \end{Bmatrix} - \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} P_i \\ P_j \end{Bmatrix} \end{aligned}$$



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Principle of minimum potential energy for a single element

$$\Pi = U + W = \frac{1}{2} \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} - \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} \frac{ql}{2} \\ q\frac{l}{2} \end{Bmatrix} - \begin{bmatrix} u_i & u_j \end{bmatrix} \begin{Bmatrix} P_i \\ P_j \end{Bmatrix}$$

$$\frac{\partial \Pi}{\partial \mathbf{u}^T} = \begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} - \begin{Bmatrix} \frac{ql}{2} \\ q\frac{l}{2} \end{Bmatrix} - \begin{Bmatrix} P_i \\ P_j \end{Bmatrix} = 0$$

$$\begin{bmatrix} \frac{EA}{l} & -\frac{EA}{l} \\ -\frac{EA}{l} & \frac{EA}{l} \end{bmatrix} \begin{Bmatrix} u_i \\ u_j \end{Bmatrix} = \begin{Bmatrix} \frac{ql}{2} + P_i \\ q\frac{l}{2} + P_j \end{Bmatrix}$$

Element equilibrium equation



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$$\mathbf{k}\mathbf{u} = \mathbf{f}$$

where

$$\mathbf{k} = \int_V (\mathbf{B}^T \mathbf{E} \mathbf{B}) dV$$

is the *element stiffness matrix*.



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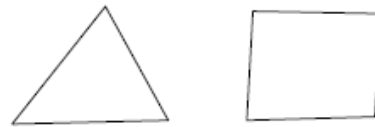


Choosing elements

- Choose the right type of element for a given problem
- Avoid elements with large aspect ratios and corner angle



Elements with Bad Shapes



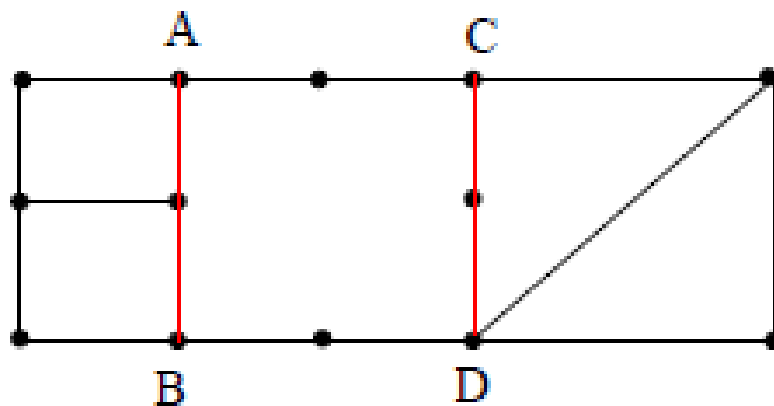
Elements with Nice Shapes

- Connect elements correctly



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Modeling and solution techniques

- Use symmetry
 - Reflective symmetry
 - Rotational (cyclic symmetry)
 - Axisymmetry
- Substructures
- Convergence of FE solutions
 - Reduce the element size
 - Increase the order of element



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Global dynamic equations

$$[\mathbf{M}]\{\ddot{\mathbf{D}}\} + [\mathbf{C}]\{\dot{\mathbf{D}}\} + [\mathbf{K}]\{\mathbf{D}\} = \{\mathbf{R}^{\text{ext}}\}$$



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Explicit integration

$$\frac{1}{\Delta t^2} [\mathbf{M}] \{\mathbf{D}\}_{n+1} = \{\mathbf{R}^{\text{ext}}\}_n - \left[\mathbf{K} - \frac{2}{\Delta t^2} \mathbf{M} + \frac{1}{\Delta t} \mathbf{C} \right] \{\mathbf{D}\}_n - \left[\frac{2}{\Delta t^2} \mathbf{M} - \frac{1}{\Delta t} \mathbf{C} \right] \{\mathbf{D}\}_{n-1}$$



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Implicit integration

$$\begin{aligned} [\mathbf{K}^{\text{eff}}] \{\mathbf{D}\}_{n+1} &= \{\mathbf{R}^{\text{ext}}\}_{n+1} + [\mathbf{M}] \left[\frac{1}{\beta \Delta t^2} \{\mathbf{D}\}_n + \frac{1}{\beta \Delta t} \{\dot{\mathbf{D}}\}_n + \left(\frac{1}{2\beta} - 1 \right) \{\ddot{\mathbf{D}}\}_n \right] \\ &+ [\mathbf{C}] \left[\frac{\gamma}{\beta \Delta t} \{\mathbf{D}\}_n + \left(\frac{\gamma}{\beta} - 1 \right) \{\dot{\mathbf{D}}\}_n + \Delta t \left(\frac{\gamma}{2\beta} - 1 \right) \{\ddot{\mathbf{D}}\}_n \right] \end{aligned}$$

where

$$[\mathbf{K}^{\text{eff}}] = \frac{1}{\beta \Delta t^2} [\mathbf{M}] + \frac{\gamma}{\beta \Delta t} [\mathbf{C}] + [\mathbf{K}]$$



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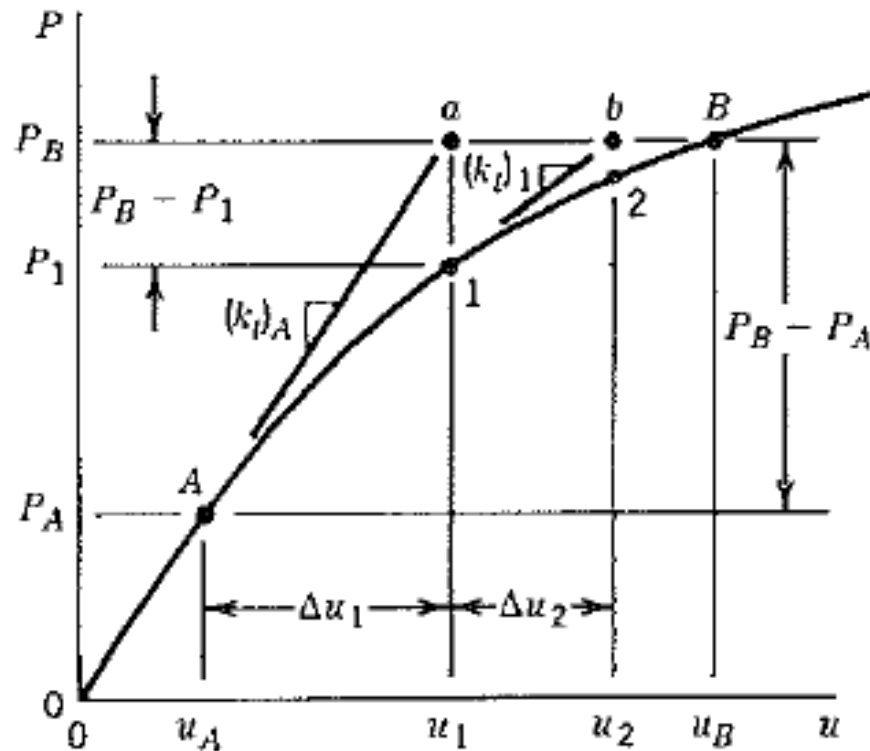


Nonlinear problems

- Material nonlinearity
- Contact nonlinearity
- Geometric nonlinearity



Newton-Raphson (N-R)



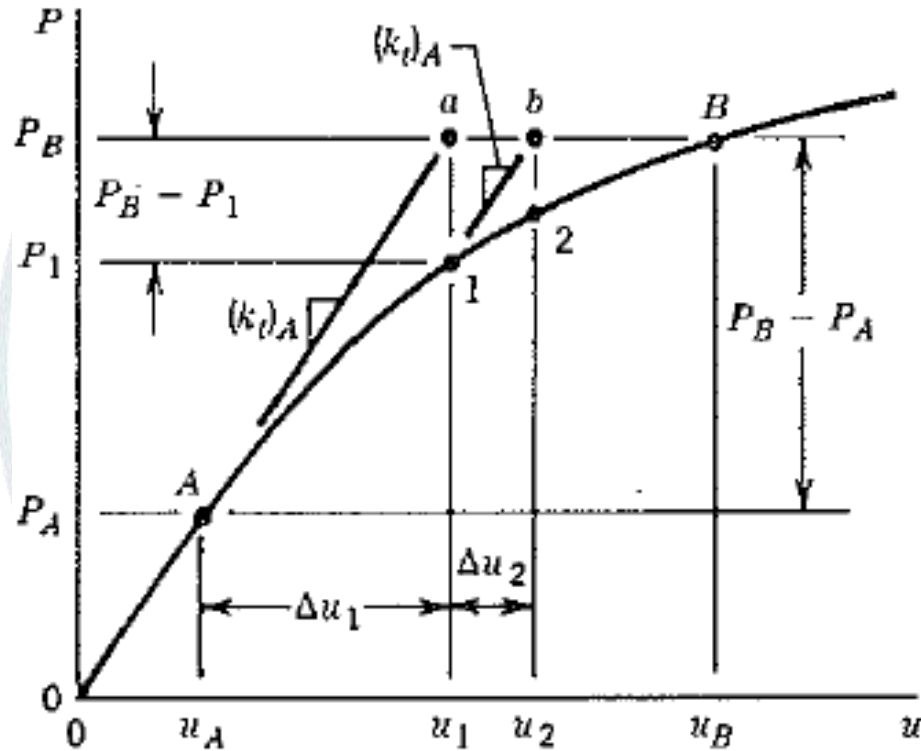


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Modified Newton-Raphson iteration



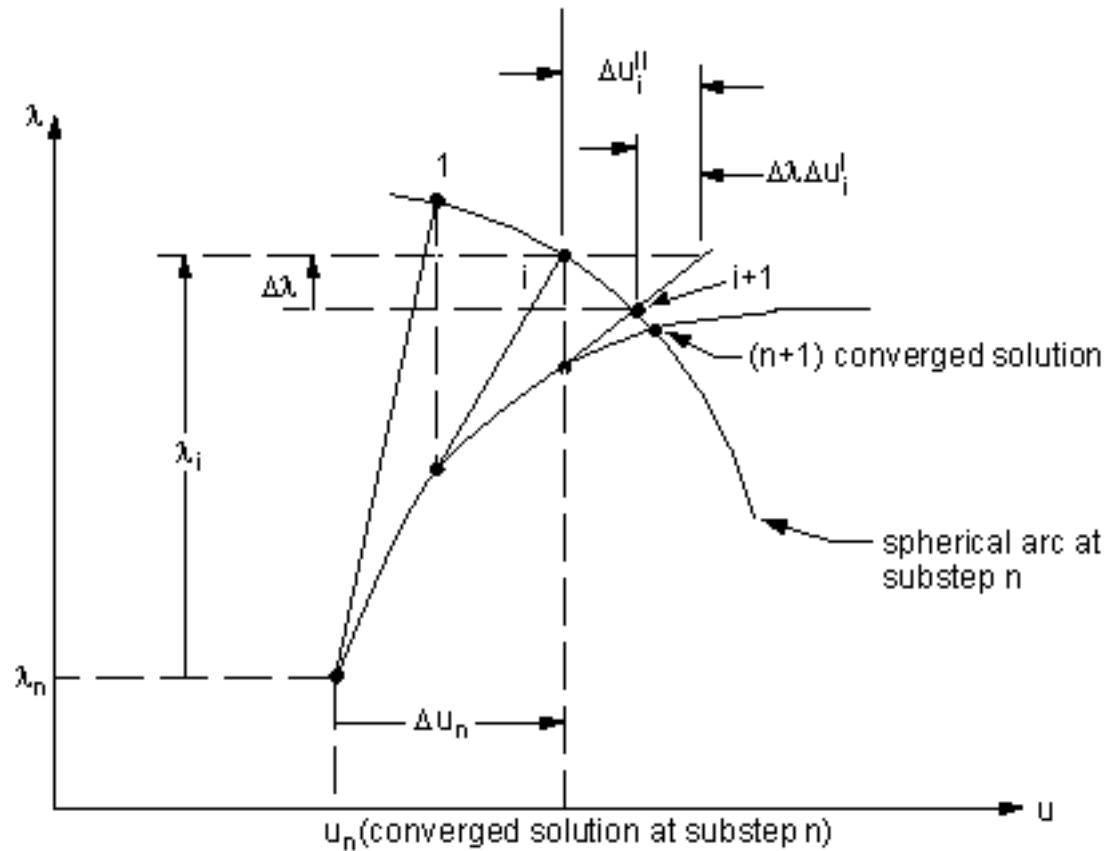


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Arc-length method





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References



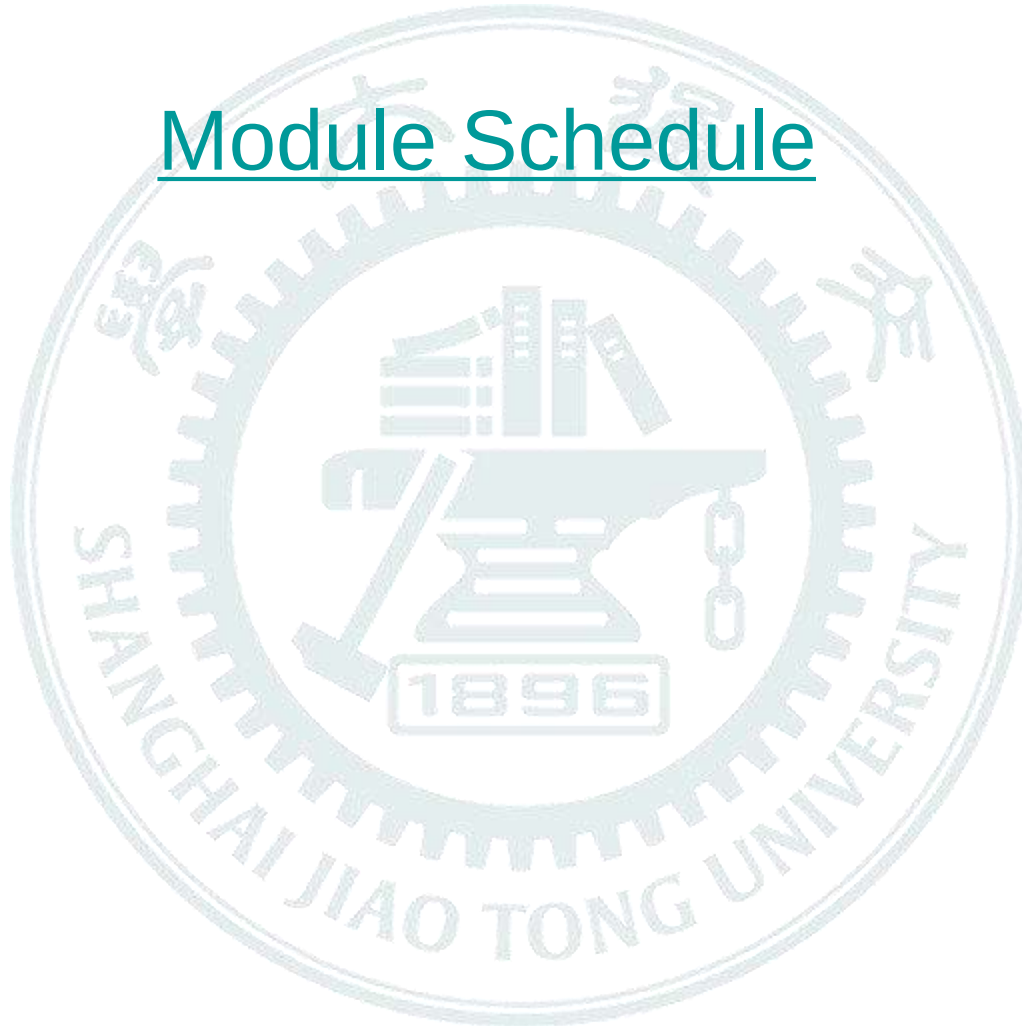
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Module Schedule





Thanks !

